

EX NO: 2a	GENERATION AND ANALYSIS OF FREQUENCY MODULATED (FM) SIGNALS USING MATLAB
DATE:	

Objective 2: To analyze the frequency spectrum of the generated FM signal using FFT and identify the carrier component, sideband frequencies, and the bandwidth of the FM signal based on Carson's rule

PROGRAM (MATLAB CODE)

```

/MATLAB Drive/experiment_3.m
1  % Calculation and Verification of Frequency Deviation
2  % and Modulation Index of FM Signals
3
4  clc;
5  clear;
6  close all;
7
8  %% Input Parameters
9  Am = 2;           % Message signal amplitude (V)
10 fm = 500;         % Message frequency (Hz)
11 Ac = 1;           % Carrier amplitude
12 fc = 10000;        % Carrier frequency (Hz)
13 kf = 1000;         % Frequency sensitivity (Hz/V)
14 fs = 100000;        % Sampling frequency (Hz)
15 T = 0.02;          % Simulation duration (s)
16
17 t = 0:1/fs:T;
18
19 %% Message Signal
20 m = Am*cos(2*pi*fm*t);
21
22 %% Frequency Deviation Calculation
23 delta_f = kf * Am;
24
25 %% Modulation Index Calculation
26 beta = delta_f / fm;
27
28 %% FM Signal Generation
29 fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
30
31 %% Plot Message Signal
32 figure;
33
34 subplot(2,1,1);
35 plot(t, m, 'LineWidth', 1.5);
36 title('Message Signal');
37 xlabel('Time (s)');
38 ylabel('Amplitude');
39 grid on;
40
41 %% Plot FM Signal
42 subplot(2,1,2);
43 plot(t, fm_signal, 'LineWidth', 1.5);
44 title('Frequency Modulated Signal');
45 xlabel('Time (s)');
46 ylabel('Amplitude');
47 grid on;
48
49 %% Spectrum Analysis
50 N = length(fm_signal);
51
52 FM_fft = fftshift(abs(fft(fm_signal)) / N);
53
54 f = (-N/2:N/2-1) * (fs/N);
55
56 figure;
57 plot(f, FM_fft, 'LineWidth', 1.5);
58 title('FM Signal Spectrum');
59 xlabel('Frequency (Hz)');
60
61 title('FM Signal Spectrum');
62 xlabel('Frequency (Hz)');
63 ylabel('Magnitude');
64 grid on;
65
66 xlim([fc-10000 fc+10000]);
67
68 %% Display Results
69 fprintf('\nFM SIGNAL PARAMETERS\n');
70 fprintf('-----\n');
71 fprintf('Carrier Frequency = %.0f Hz\n', fc);
72 fprintf('Message Frequency = %.0f Hz\n', fm);
73 fprintf('Message Amplitude = %.2f V\n', Am);
74 fprintf('Frequency Sensitivity = %.0f Hz/V\n', kf);
75
76 fprintf('\nCALCULATED VALUES\n');
77 fprintf('-----\n');
78 fprintf('Frequency Deviation (Delta f) = %.0f Hz\n', delta_f);
79 fprintf('Modulation Index (beta) = %.2f\n', beta);
80
81 %% Carson's Rule Bandwidth
82 BW = 2 * (delta_f + fm);
83
84 fprintf('Bandwidth (Carson Rule) = %.0f Hz\n', BW);

```

Procedure

1. Open MATLAB and create a new script file.
2. Clear the command window, workspace variables, and close all existing figure windows using:

- ```

 clc;
 Clear;
 close all;

```
- Define the message signal parameters:
    - Message amplitude  $A_m=2V$
    - Message frequency  $f_m=500Hz$
  - Define the carrier signal parameters:
    - Carrier amplitude  $A_c=1V$
    - Carrier frequency  $f_c=10000Hz$
  - Set the frequency sensitivity constant  $k_f=1000Hz/V$ , sampling frequency  $f_s=100000Hz$ , and simulation duration  $T=0.02s$ .
  - Generate the time vector using:
 
$$t = 0:1/f_s:T;$$
  - Generate the message signal using:
 
$$m = A_m \cdot \cos(2\pi \cdot f_m \cdot t);$$
  - Calculate the frequency deviation of the FM signal using:
 
$$\Delta f = k_f \cdot A_m$$
  - Calculate the modulation index using:
 
$$\beta = \Delta f / f_m$$
  - Generate the FM signal using:
 
$$fm\_signal = A_c \cdot \cos(2\pi \cdot f_c \cdot t + \beta \cdot \sin(2\pi \cdot f_m \cdot t));$$
  - Plot the message signal in the time domain and observe its waveform.
  - Plot the generated FM signal and observe that the carrier frequency varies according to the message signal while the amplitude remains constant.
  - Determine the length of the FM signal:
 
$$N = \text{length}(fm\_signal);$$
  - Compute the Fast Fourier Transform (FFT) of the FM signal:
 
$$FM\_fft = \text{fft}(fm\_signal);$$
  - Shift the spectrum to center the zero-frequency component and normalize it:
 
$$FM\_fft = \text{fftshift}(\text{abs}(FM\_fft)/N);$$
  - Generate the frequency axis corresponding to the FFT output:
 
$$f = (-N/2:N/2-1) \cdot (f_s/N);$$
  - Plot the frequency spectrum of the FM signal and observe the carrier component and sidebands.
  - Restrict the frequency axis around the carrier frequency for better visualization:
 
$$\text{xlim}([f_c-10000 \ f_c+10000]);$$
  - Calculate the bandwidth of the FM signal using Carson's Rule:
 
$$BW = 2(\Delta f + f_m)$$
  - Display the carrier frequency, message frequency, frequency deviation, modulation index, and bandwidth in the MATLAB command window.

21. Execute the program and verify that the calculated frequency deviation and modulation index match the values used in generating the FM signal

## OUTPUT

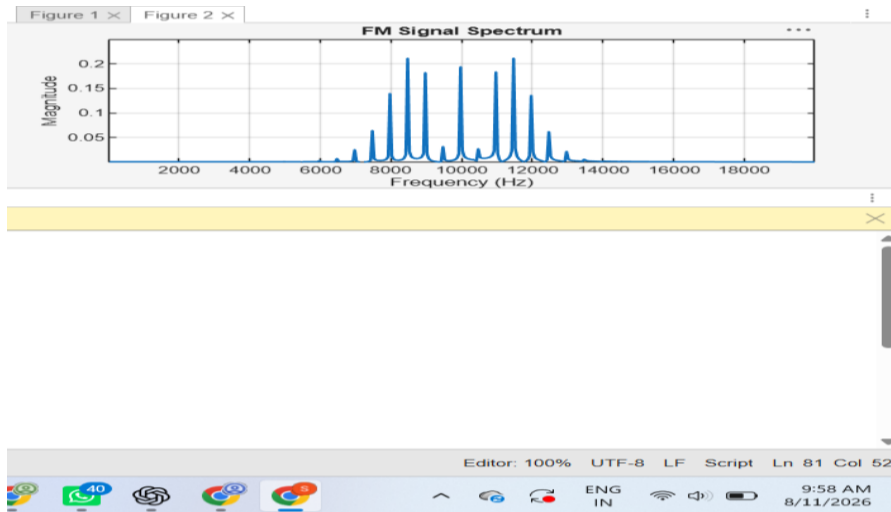
```
FM SIGNAL PARAMETERS

Carrier Frequency = 10000 Hz
Message Frequency = 500 Hz
Message Amplitude = 2.00 V
Frequency Sensitivity = 1000 Hz/V

CALCULATED VALUES

Frequency Deviation (Delta f) = 2000 Hz
Modulation Index (beta) = 4.00
Bandwidth (Carson Rule) = 5000 Hz
>>
```

## OUTPUT (GRAPH)



## Result

- The time-domain waveform was observed, and the frequency spectrum obtained using FFT has been plotted.
- Also, calculated the frequency deviation and bandwidth using Carson's rule.